

# AIRLINE PASSENGER

## SATISFACTION ANALYSIS

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# DATASET OVERVIEW & DATA VISUALIZATION

**129,880**

Total Records

**23**

Feature Columns

**393**

Missing Values (Arrival Delay only)

**0**

Duplicate Rows

**43.4%**

Satisfied Passengers

## WHY DATA VISUALIZATION?



### Univariate

Distributions of single variables (histograms, count plots)



### Bivariate

Relationships between two variables (bar, KDE, boxplots)



### Multivariate

Correlations across all features (heatmaps)



### Feature Importance

Which variables drive the target (bar chart)

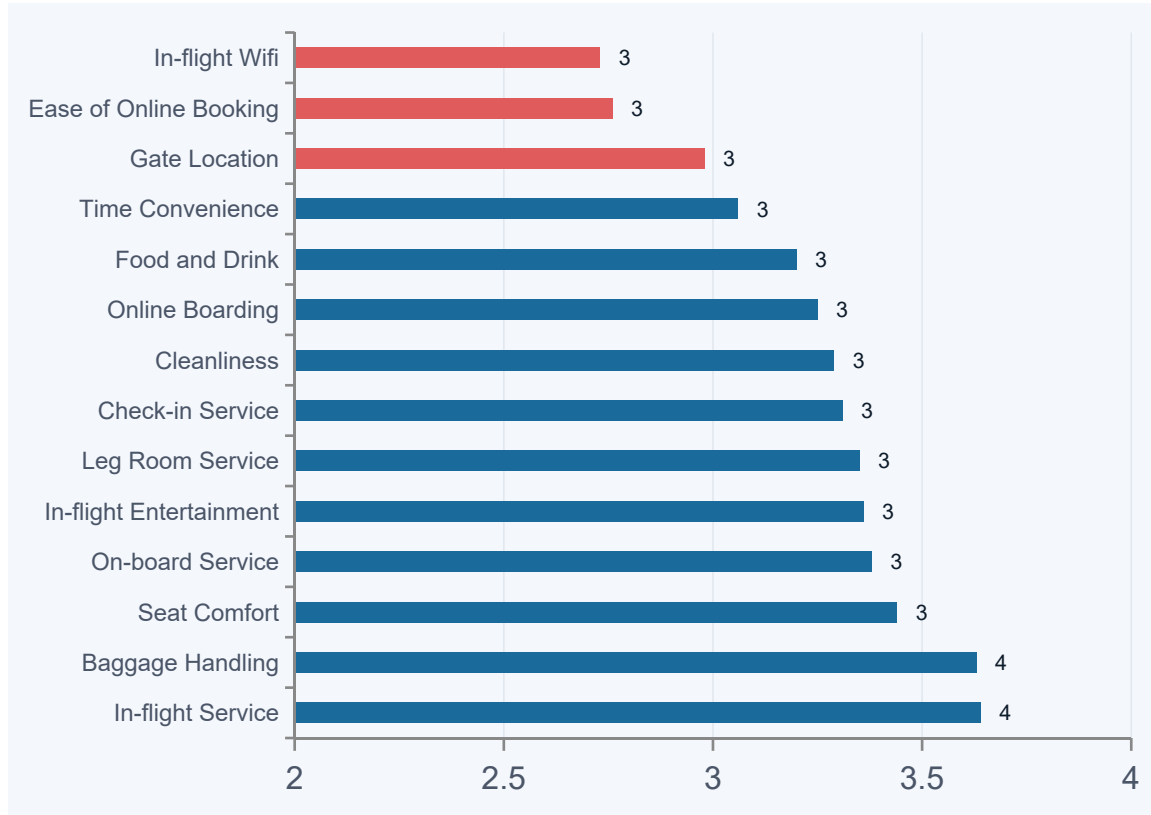


### Outlier Detection

Extreme values via IQR & boxplots

# SERVICE RATINGS & KEY SATISFACTION DRIVERS

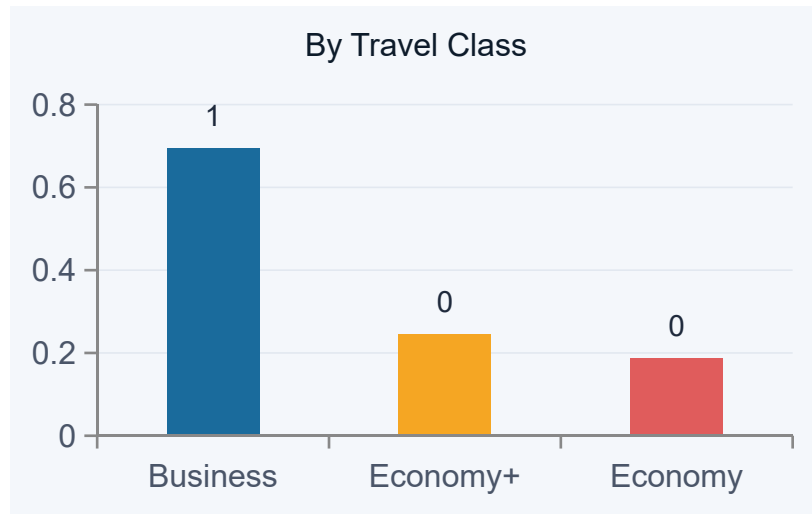
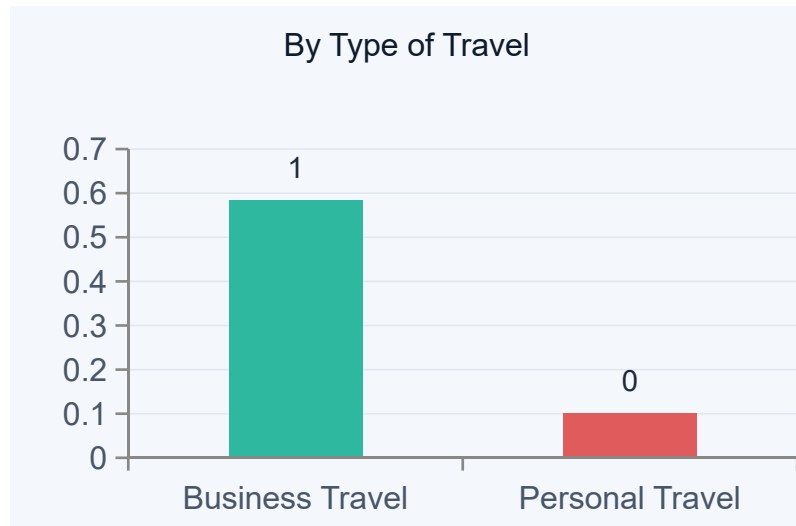
*Univariate Analysis — average rating per service attribute (scale 1–5), visualised as a horizontal bar chart*



*Red bars = digital pain points (lowest ratings)*

# EDA: PASSENGER SEGMENTS & SATISFACTION RATES

*Bivariate Analysis — comparing a categorical variable against the binary target (Satisfaction) using grouped bar charts*



Business Travelers are 5.8x more satisfied as then personal travelers (58.4% to 10.1%)

Business class leads at 69.4%; Economy sits at just 18.8%

Returning customers (47.8%) are 2x more satisfied than first-timers (24.0%)

# NUMERICAL & SERVICE CORRELATION HEATMAP

Correlation between numerical and service-related features (Red = Positive Correlation, Blue = Negative Correlation)

## KEY INSIGHTS



Departure Delay and Arrival Delay are **highly correlated (0.96)**.



Online Boarding has a strong positive correlation with Satisfaction (0.50).



In-flight Wifi and Ease of Online Booking are **highly correlated (0.71)**.



Cabin experience features (Seat Comfort, Cleanliness, Food & Drink, Entertainment) show **strong inter-correlation**.



Satisfaction shows **moderate positive correlation** with key service features.

## CORRELATION SCALE



|                                        | Age   | Flight Distance | Departure Delay | Arrival Delay | Departure and Arrival Time Convenience | Ease of Online Booking | Check-in Service | Online Boarding | Gate Location | On-board Service | Seat Comfort | Leg Room Service | Cleanliness | Food and Drink | In-flight Service | In-flight Wifi Service | In-flight Entertainment | Baggage Handling | Satisfaction |
|----------------------------------------|-------|-----------------|-----------------|---------------|----------------------------------------|------------------------|------------------|-----------------|---------------|------------------|--------------|------------------|-------------|----------------|-------------------|------------------------|-------------------------|------------------|--------------|
| Age                                    | 1.00  | 0.10            | -0.01           | -0.01         | 0.04                                   | 0.02                   | 0.03             | 0.21            | -0.00         | 0.06             | 0.16         | 0.04             | 0.05        | 0.02           | -0.05             | 0.02                   | 0.07                    | -0.05            | 0.13         |
| Flight Distance                        | 0.10  | 1.00            | 0.00            | -0.00         | -0.02                                  | 0.07                   | 0.07             | 0.21            | 0.01          | 0.11             | 0.16         | 0.13             | 0.10        | 0.06           | 0.06              | 0.01                   | 0.13                    | 0.06             | 0.30         |
| Departure Delay                        | -0.01 | 0.00            | 1.00            | 0.96          | 0.00                                   | -0.01                  | -0.02            | -0.02           | 0.01          | -0.03            | -0.03        | 0.01             | -0.01       | -0.03          | -0.05             | -0.02                  | -0.03                   | -0.00            | -0.05        |
| Arrival Delay                          | -0.01 | -0.00           | 0.96            | 1.00          | -0.00                                  | -0.01                  | -0.02            | -0.02           | 0.01          | -0.03            | -0.03        | 0.01             | -0.02       | -0.03          | -0.06             | -0.02                  | -0.03                   | -0.01            | -0.06        |
| Departure and Arrival Time Convenience | 0.04  | -0.02           | 0.00            | -0.00         | 1.00                                   | 0.44                   | 0.09             | 0.07            | 0.45          | 0.07             | 0.01         | 0.01             | 0.01        | 0.00           | 0.07              | 0.34                   | -0.01                   | 0.07             | -0.05        |
| Ease of Online Booking                 | 0.02  | 0.07            | -0.01           | -0.01         | 0.44                                   | 1.00                   | 0.01             | 0.40            | 0.46          | 0.04             | 0.03         | 0.11             | 0.02        | 0.03           | 0.04              | 0.71                   | 0.05                    | 0.04             | 0.17         |
| Check-in Service                       | 0.03  | 0.07            | -0.02           | -0.02         | 0.09                                   | 0.01                   | 1.00             | 0.20            | -0.04         | 0.24             | 0.19         | 0.15             | 0.18        | 0.09           | 0.24              | 0.04                   | 0.12                    | 0.23             | 0.24         |
| Online Boarding                        | 0.21  | 0.21            | -0.02           | -0.02         | 0.07                                   | 0.40                   | 0.20             | 1.00            | 0.00          | 0.15             | 0.42         | 0.12             | 0.33        | 0.23           | 0.07              | 0.46                   | 0.28                    | 0.08             | 0.50         |
| Gate Location                          | -0.00 | 0.01            | 0.01            | 0.01          | 0.45                                   | 0.46                   | -0.04            | 0.00            | 1.00          | -0.03            | 0.00         | -0.01            | -0.01       | -0.00          | 0.00              | 0.34                   | 0.00                    | 0.00             | -0.00        |
| On-board Service                       | 0.06  | 0.11            | -0.03           | -0.03         | 0.07                                   | 0.04                   | 0.24             | 0.15            | -0.03         | 1.00             | 0.13         | 0.36             | 0.12        | 0.06           | 0.55              | 0.12                   | 0.42                    | 0.52             | 0.32         |
| Seat Comfort                           | 0.16  | 0.16            | -0.03           | -0.03         | 0.01                                   | 0.03                   | 0.19             | 0.42            | 0.00          | 0.13             | 1.00         | 0.10             | 0.68        | 0.58           | 0.07              | 0.12                   | 0.61                    | 0.07             | 0.35         |
| Leg Room Service                       | 0.04  | 0.13            | 0.01            | 0.01          | 0.01                                   | 0.11                   | 0.15             | 0.12            | -0.01         | 0.36             | 0.10         | 1.00             | 0.10        | 0.03           | 0.37              | 0.16                   | 0.30                    | 0.37             | 0.31         |
| Cleanliness                            | 0.05  | 0.10            | -0.01           | -0.02         | 0.01                                   | 0.02                   | 0.18             | 0.33            | -0.01         | 0.12             | 0.68         | 0.10             | 1.00        | 0.66           | 0.09              | 0.13                   | 0.69                    | 0.10             | 0.31         |
| Food and Drink                         | 0.02  | 0.06            | -0.03           | -0.03         | 0.00                                   | 0.03                   | 0.09             | 0.23            | -0.00         | 0.06             | 0.58         | 0.03             | 0.66        | 1.00           | 0.04              | 0.13                   | 0.62                    | 0.04             | 0.21         |
| In-flight Service                      | -0.05 | 0.06            | -0.05           | -0.06         | 0.07                                   | 0.04                   | 0.24             | 0.07            | 0.00          | 0.55             | 0.07         | 0.37             | 0.09        | 0.04           | 1.00              | 0.11                   | 0.41                    | 0.63             | 0.24         |
| In-flight Wifi Service                 | 0.02  | 0.01            | -0.02           | -0.02         | 0.34                                   | 0.71                   | 0.04             | 0.46            | 0.34          | 0.12             | 0.12         | 0.16             | 0.13        | 0.13           | 0.11              | 1.00                   | 0.21                    | 0.12             | 0.28         |
| In-flight Entertainment                | 0.07  | 0.13            | -0.03           | -0.03         | -0.01                                  | 0.05                   | 0.12             | 0.28            | 0.00          | 0.42             | 0.61         | 0.30             | 0.69        | 0.62           | 0.41              | 0.21                   | 1.00                    | 0.38             | 0.40         |
| Baggage Handling                       | -0.05 | 0.06            | -0.00           | -0.01         | 0.07                                   | 0.04                   | 0.23             | 0.08            | 0.00          | 0.52             | 0.07         | 0.37             | 0.10        | 0.04           | 0.63              | 0.12                   | 0.38                    | 1.00             | 0.25         |
| Satisfaction                           | 0.13  | 0.30            | -0.05           | -0.06         | -0.05                                  | 0.17                   | 0.24             | 0.50            | -0.00         | 0.32             | 0.35         | 0.31             | 0.31        | 0.21           | 0.24              | 0.28                   | 0.40                    | 0.25             | 1.00         |



This heatmap helps identify **strong relationships, multicollinearity and key drivers** of passenger satisfaction.

**THANK YOU**

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